

EXERCISE NO.1

Code:

Main Funtion:

Code:

```
public class EntryCall {
    public void complexNumberExercise() {
        ComplexNumber[] arr = new ComplexNumber[5];
        // Loop runs 5 times
        for(int iTem = 0; iTem < 5; iTem++) {
            arr[iTem] = new ComplexNumber();
            System.out.println("Enter number1:");
            int num1 = ConsoleInput.getInt();

            System.out.println("Enter number2:");
            int num2 = ConsoleInput.getInt();

            arr[iTem].setNumber1(num1);
            arr[iTem].setNumber2(num2);
        }
        System.out.println("Results:");
        for(int iTem = 0; iTem < 5; iTem++) {
            int result = arr[iTem].computeComplexNumber();
            System.out.println(result);
        }
    }

    public static void main(String[] args) {
//        FORLOOP Exercise
        Loop Fobj = new Loop();
        Fobj.displayOdd();
        Fobj.displayEven();
        Fobj.displaySeventh();
        Fobj.star1();
        Fobj.star2();
        Fobj.star3();
        Fobj.star4();
        Fobj.star5();

//        ARRAY Exercise
        EntryCall obj = new EntryCall();

        Array obj2 = new Array();
        System.out.println("Enter size:");
```

```
int size = ConsoleInput.getInt();
```

```
obj2.sortArray(size);
```

```
obj2.SumValue(size);
```

```
obj2.Average(size);
```

```
obj2.Copy(size);
```

```
obj2.MaxMin(size);
```

```
obj2.rev(size);
```

```
obj2.Duplicate(size);
```

```
obj.complexNumberExercise();
```

```
System.out.println("Enter size of first array:");
```

```
int size1 = ConsoleInput.getInt();
```

```
System.out.println("Enter size of second array:");
```

```
int size2 = ConsoleInput.getInt();
```

```
obj2.CommonElements(size1, size2);
```

```
// OOP Exercise
```

```
Student.exercise1();
```

```
Triangle.exercise2();
```

```
Rectangle.exercise3();
```

```
Complex.exercise4();
```

```
Employee.exercise5();
```

```
Employee1.exercise6();
```

```
}
```

```
}
```

FORLOOP SOLUTION:

Loop.java:

Code:

//Question 1 Solution :

```
public class Loop {  
    public void displayOdd() {  
        for(int iTem = 1; iTem <=1000; iTem++) {  
            if(iTem % 2 != 0)  
                System.out.print(iTem + " ");  
        }  
    }  
}
```

//Question 2 Solution :

```
    public void displayEven() {  
        for(int iTem = 1; iTem <=500; iTem++) {  
            if(iTem % 2 == 0)  
                System.out.print(iTem + " ");  
        }  
    }  
}
```

//Question 3 Solution:

```
    public void displaySeventh() {  
        for(int iTem = 7; iTem <=200; iTem = iTem + 7)  
            System.out.print(iTem + " ");  
    }  
}
```

//Question 4 Solution:

```
    public void star1() {  
        int num = 5;  
        for(int iTem = 1; iTem <= num; iTem++) {  
            for(int jTem = 1; jTem <= iTem; jTem++) {  
                System.out.print("*");  
            }  
            System.out.println();  
        }  
    }  
}
```

//Question 5 Solution:

```
    public void star2() {  
        int num = 5;  
        for(int iTem = 1; iTem <= num; iTem++) {  
            for(int jTem = 1; jTem <= num - iTem + 1; jTem++) {  
                System.out.print("*");  
            }  
        }  
    }  
}
```

```
        System.out.println();
    }
}
```

//Question 6 Solution :

```
public void star3() {
    int num = 1;
    for(int iTem = 1; iTem <= 4; iTem++) {
        for(int jTem = 1; jTem <= iTem; jTem++) {
            System.out.print(num+" ");
            num += 1;
        }
        System.out.println();
    }
}
```

//Question 7 Solution:

```
public void star4() {
    int num = 6;
    for(int iTem = 1; iTem <= num; iTem++) {
        for(int jTem = 1; jTem <= iTem; jTem++) {
            System.out.print(jTem+" ");
        }
        System.out.println();
    }
}
```

//Question 8 Solution :

```
public void star5() {
    int num = 6;
    for(int iTem = 1; iTem <= num; iTem++) {
        for(int jTem = 1; jTem <= num - iTem + 1; jTem++) {
            System.out.print(jTem+" ");
        }
        System.out.println();
    }
}

}
```

ARRAY SOLUTION:

Array.java:

Code:

```
import java.util.Arrays;
public class Array {
```

//Question 1 Solution :

```
    public void sortArray(int size) {
        int[] arr = new int[size];

        for(int iTem = 0; iTem < size; iTem++) {
            System.out.println("Enter Elements:");
            arr[iTem] = ConsoleInput.getInt();
        }

        for(int iTem = 0; iTem < size - 1; iTem++) {
            for(int jTem = 0; jTem < size - iTem - 1; jTem++) {
                if(arr[jTem] > arr[jTem + 1]) {

                    int temp = arr[jTem];
                    arr[jTem] = arr[jTem + 1];
                    arr[jTem + 1] = temp;
                }
            }
        }

        System.out.println("Sorted Array:");
        for(int print:arr) {
            System.out.print(print+" ");
        }
    }
```

//Question 2 Solution :

```
    public void SumValue(int size) {
        int[] arr = new int[size];
        for(int iTem = 0; iTem < size; iTem++) {
            System.out.println("Enter Elements:");
            arr[iTem] = ConsoleInput.getInt();
        }
        int sum = 0;
        for(int iTem = 0; iTem < size; iTem++) {
            sum += arr[iTem];
        }
        System.out.println(sum);
    }
```

```
}
```

//Question 3 Solution :

```
public void Average(int size) {  
    int[] arr = new int[size];  
    for(int iTem = 0; iTem < size; iTem++) {  
        System.out.println("Enter Elements:");  
        arr[iTem] = ConsoleInput.getInt();  
    }  
    int sum = 0;  
    for(int iTem = 0; iTem < size; iTem++) {  
        sum += arr[iTem];  
    }  
    System.out.println((double)sum/size);  
}
```

//Question 4 Solution :

```
public void Copy(int size) {  
    int[] arr = new int[size];  
    for(int iTem = 0; iTem < size; iTem++) {  
        System.out.println("Enter Elements:");  
        arr[iTem] = ConsoleInput.getInt();  
    }  
    int[] answer = new int[size];  
  
    for(int iTem = 0; iTem < size; iTem++) {  
        answer[iTem] = arr[iTem];  
    }  
    System.out.print("Copying Elements are: ");  
    for(int print:answer) {  
        System.out.print(print+" ");  
    }  
}
```

//Question 5 Solution :

```
public void MaxMin(int size) {  
    int[] arr = new int[size];  
    for(int iTem = 0; iTem < size; iTem++) {  
        System.out.println("Enter Elements:");  
        arr[iTem] = ConsoleInput.getInt();  
    }  
  
    int Max = Integer.MIN_VALUE, Min = Integer.MAX_VALUE;  
    for(int iTem = 0; iTem < size; iTem++) {
```

```

        if(Max < arr[iTem]) Max = arr[iTem];
        if(Min > arr[iTem]) Min = arr[iTem];
    }
    System.out.println("Max Vaule is:"+ Max);
    System.out.println("Min Vaule is:"+ Min);
}

```

//Question 6 Solution :

```

public void rev(int size) {
    int[] arr = new int[size];
    for(int iTem = 0; iTem < size; iTem++) {
        System.out.println("Enter Elements:");
        arr[iTem] = ConsoleInput.getInt();
    }
    int[] answer = new int[size];
    for(int iTem = 0; iTem < size; iTem++) {
        answer[iTem] = arr[size - iTem - 1];
    }
    for(int print:answer) {
        System.out.print(print+" ");
    }
}

```

//Question 7 Solution :

```

public void Duplicate(int size) {
    int[] arr = new int[size];
    for(int iTem = 0; iTem < size; iTem++) {
        System.out.println("Enter Elements:");
        arr[iTem] = ConsoleInput.getInt();
    }
    Arrays.sort(arr);
    for(int iTem = 0; iTem < size-1 ; iTem++) {
        if(arr[iTem] == arr[ iTem+ 1]) {
            System.out.println(arr[iTem]);
        }
    }
}

```

//Question 8 Solution :

```

// Find common elements between two arrays
public void CommonElements(int size1, int size2) {
    int[] arr1 = new int[size1];
    int[] arr2 = new int[size2];
    // Take elements for first array

```

```

        System.out.println("Enter elements for first array:");

        for(int iTem = 0; iTem < size1; iTem++) {

            arr1[iTem] = ConsoleInput.getInt();
        }
        System.out.println("Enter elements for second array:");

        for(int iTem = 0; iTem < size2; iTem++) {
            arr2[iTem] = ConsoleInput.getInt();
        }
        System.out.println("Common Elements:");
        for(int iTem = 0; iTem < size1; iTem++) {
            for(int jTem = 0; jTem < size2; jTem++) {
                // If both values are same
                if(arr1[iTem] == arr2[jTem]) {
                    System.out.println(arr1[iTem]);
                    break;
                }
            }
        }
    }
}

```

ComplexNumber:

```

public class ComplexNumber {

    private int number1;
    private int number2;

    public void setNumber1(int number1) {
        this.number1 = number1;
    }

    public void setNumber2(int number2) {
        this.number2 = number2;
    }

    public int getNumber1() {
        return number1;
    }

    public int getNumber2() {
        return number2;
    }
}

```

```

    }

    public int computeComplexNumber() {
        return number1 * number2;
    }
}

```

CLASS SOLUTION:

Student.java:

//Question 1 Solution :

Code:

```

public class Student {
    String name;
    int roll_no;
    String phone_no;
    String address;

    Student(String name, int roll_no, String phone_no, String address) {
        this.name = name;
        this.roll_no = roll_no;
        this.phone_no = phone_no;
        this.address = address;
    }

    public static void exercise1a() {

        Student s1 = new Student("John", 2, "", "");

        System.out.println("Name: " + s1.name);
        System.out.println("Roll No: " + s1.roll_no);

        System.out.println("Name: " + s1.name);
        System.out.println("Roll No: " + s1.roll_no);
    }

    public static void exercise1b() {
        Student s1 = new Student("Sam", 1, "9876543210", "Mumbai");

        Student s2 = new Student("John", 2, "9876501234", "Pune");
        System.out.println("Student 1");
        System.out.println("Name: " + s1.name);
        System.out.println("Roll No: " + s1.roll_no);
    }
}

```

```

        System.out.println("Phone No: " + s1.phone_no);
        System.out.println("Address: " + s1.address);

        System.out.println();

        System.out.println("Student 2");
        System.out.println("Name: " + s2.name);
        System.out.println("Roll No: " + s2.roll_no);
        System.out.println("Phone No: " + s2.phone_no);
        System.out.println("Address: " + s2.address);
    }

    public static void exercise1() {
        exercise1a();
        System.out.println();
        exercise1b();
    }
}

```

Rectangle.java:

Code:

//Question 2 Solution :

```

public class Rectangle {
    private int length;
    private int breadth;
    public Rectangle(int length, int breadth) {

        this.length = length;
        this.breadth = breadth;
    }

    // Calculate area
    public int area() {
        return length * breadth;
    }

    public static void exercise3() {
        Rectangle r1 = new Rectangle(4, 5);
        Rectangle r2 = new Rectangle(5, 8);
        System.out.println("Rectangle 1 Area: " + r1.area());
        System.out.println("Rectangle 2 Area: " + r2.area());
    }
}

```

Triangle.java:**Code:****//Question 3 Solution :**

```
public class Triangle {

    private int side1;
    private int side2;
    private int side3;
    public Triangle(int side1, int side2, int side3) {

        this.side1 = side1;
        this.side2 = side2;
        this.side3 = side3;
    }
    // Calculate perimeter
    public int perimeter() {
        return side1 + side2 + side3;
    }
    // Calculate area
    public double area() {
        double s = perimeter() / 2.0;
        return Math.sqrt(s * (s - side1) *(s - side2) *(s - side3));
    }

    public static void exercise2() {
        Triangle t = new Triangle(3, 4, 5);
        System.out.println("Area: " + t.area());
        System.out.println("Perimeter: " + t.perimeter());
    }
}
```

Complex.Java:**Code:****//Question 4 Solution :**

```
public class Complex {
    private int real;
    private int imaginary;

    public Complex(int real, int imaginary) {
        this.real = real;
        this.imaginary = imaginary;
    }
}
```

```

public Complex sum(Complex c) {
    return new Complex(real + c.real,imaginary + c.imaginary
    );
}

public Complex difference(Complex c) {
    return new Complex(real - c.real,imaginary - c.imaginary
    );
}

// Product of two complex numbers
public Complex product(Complex c) {
    int r = (real * c.real) - (imaginary * c.imaginary);
    int i = (real * c.imaginary) + (imaginary * c.real);
    return new Complex(r, i);
}

public void display() {
    System.out.println(real + " + " + imaginary + "i");
}

public static void exercise4() {
    System.out.println("Enter real part of first number:");
    int real1 = ConsoleInput.getInt();
    System.out.println("Enter imaginary part of first number:");
    int imaginary1 = ConsoleInput.getInt();

    System.out.println("Enter real part of second number:");
    int real2 = ConsoleInput.getInt();

    System.out.println("Enter imaginary part of second number:");
    int imaginary2 = ConsoleInput.getInt();

    Complex c1 = new Complex(real1, imaginary1);
    Complex c2 = new Complex(real2, imaginary2);

    System.out.println("Sum:");
    c1.sum(c2).display();
    System.out.println("Difference:");
    c1.difference(c2).display();
    System.out.println("Product:");
    c1.product(c2).display();
}
}

```

Employee.java:

Code:

//Question 5 Solution :

```
public class Employee{
    String name;
    int yearOfJoining;
    double salary;
    String address;

    public Employee(String name,int yearOfJoining,double salary,String address) {
        this.name = name;
        this.yearOfJoining = yearOfJoining;
        this.salary = salary;
        this.address = address;
    }
    public static void exercise5() {
        Employee e1 = new Employee("Robert",1994,50000,"64C-WallsStreat");
        Employee e2 = new Employee("Sam",2000,45000,"68D-WallsStreat");
        Employee e3 = new Employee("John",1999,55000,"26B-WallsStreat");
        System.out.println("Name\tYear of Joining\tSalary\tAddress");
        System.out.println(e1.name + "\t\t" +e1.yearOfJoining + "\t\t" +e1.salary + "\t" +e1.address);
        System.out.println(e2.name + "\t\t" +e2.yearOfJoining + "\t\t" +e2.salary + "\t" +e2.address);
        System.out.println(e3.name + "\t\t" +e3.yearOfJoining + "\t\t" +e3.salary + "\t" +e3.address);
    }
}
```

Employee1.java:**Code:****//Question 6 Solution :**

```
public class Employee1 {

    private double salary;
    private int hours;

    public void getInfo(double salary, int hours) {
        this.salary = salary;
        this.hours = hours;
    }

    public void addSal() {
        if(salary < 500) {
            salary = salary + 10;
        }
    }

    public void addWork() {
        if(hours > 6) {
            salary = salary + 5;
        }
    }

    public static void exercise6() {
        Employee1 e = new Employee1();
        System.out.println("Enter salary:");
        double salary = ConsoleInput.getInt();

        System.out.println("Enter hours of work per day:");
        int hours = ConsoleInput.getInt();

        e.getInfo(salary, hours);
        e.addSal();
        e.addWork();
        System.out.println("Final Salary: $" + e.salary);
    }
}
```